Master's Thesis

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From Post to Temporal Causality: Constructing Symptom Temporal Causal Graphs Using Large Language Models

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Joonhyung Kwak

A Master's Thesis Submitted to the Department of Artificial Intelligence and the Graduate School of Sungkyunkwan University in partial fulfillment of the requirements for the degree of Master of Science in Engineering

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Abstract

From Post to Temporal Causality: Constructing Symptom Temporal Causal Graphs Using Large Language Models

Mental health has emerged as a critical research domain, with significant advancements in artificial intelligence (AI) and natural language processing (NLP) paving the way for improved detection and intervention strategies. Current approaches leveraging causal methods, such as Average Treatment Effect (ATE) and PC (Peter–Clark) Algorithm, face substantial limitations in capturing the intricate temporal dynamics embedded within textual data and external behavioral contexts. These limitations compromise the explanatory power and clinical utility of mental health assessments, particularly for complex conditions like depression. To address this, our research introduces a novel approach integrating Large Language Model (LLM) with causal discovery techniques designed specifically to capture temporal nuances within and across textual data. By employing LLM-driven temporal modeling, we extract temporal-causal markers, construct temporal-aware causal graphs, and significantly enhance the explainability of building a causal graph. The effectiveness of our proposed method is rigorously compared against the conventional ATE and PC Algorithm methods. Our findings demonstrate improved predictive performance and deeper explanatory capabilities, thereby providing valuable support for clinical decision-making and enhancing the effectiveness of digital mental health interventions.

Keywords : Causal Discovery, Mental Health, Temporal Phrase Embedding, Social Media Analysis, Explainable

Chapter 1. INTRODUCTION

Understanding the causal dynamics of mental health symptoms is central to designing effective early detection and intervention strategies. Social media has emerged as a rich, real-time source of mental health signals, enabling researchers to extract symptom trajectories from user-generated text using NLP models such as MentalBERT [1]. This has led to the adaptation of causal inference methods, particularly Average Treatment Effect (ATE)-based approaches [2, 3, 4], to assess the impact of behavioral or linguistic variables on depression risk.

While ATE-based methods offer statistical interpretability by summarizing treatment effects into numeric estimates, they are inherently limited in capturing the complex nature of psychiatric symptom development. Since ATE frameworks estimate scalar average effects between predefined treatment–outcome pairs, they cannot reflect how symptoms evolve sequentially, influence one another, or vary across individuals. This restricts their utility in mental health contexts, where symptoms often co-occur, recur, and form interdependent trajectories. Unlike ATE, graph-based methods such as the PC (Peter–Clark) algorithm [5] aim to uncover broader causal structures by modeling the directional relationships among variables. Such structure-learning approaches are particularly valuable for representing symptom-to-symptom dynamics, making them conceptually well-suited for modeling mental health as a network of interacting conditions.

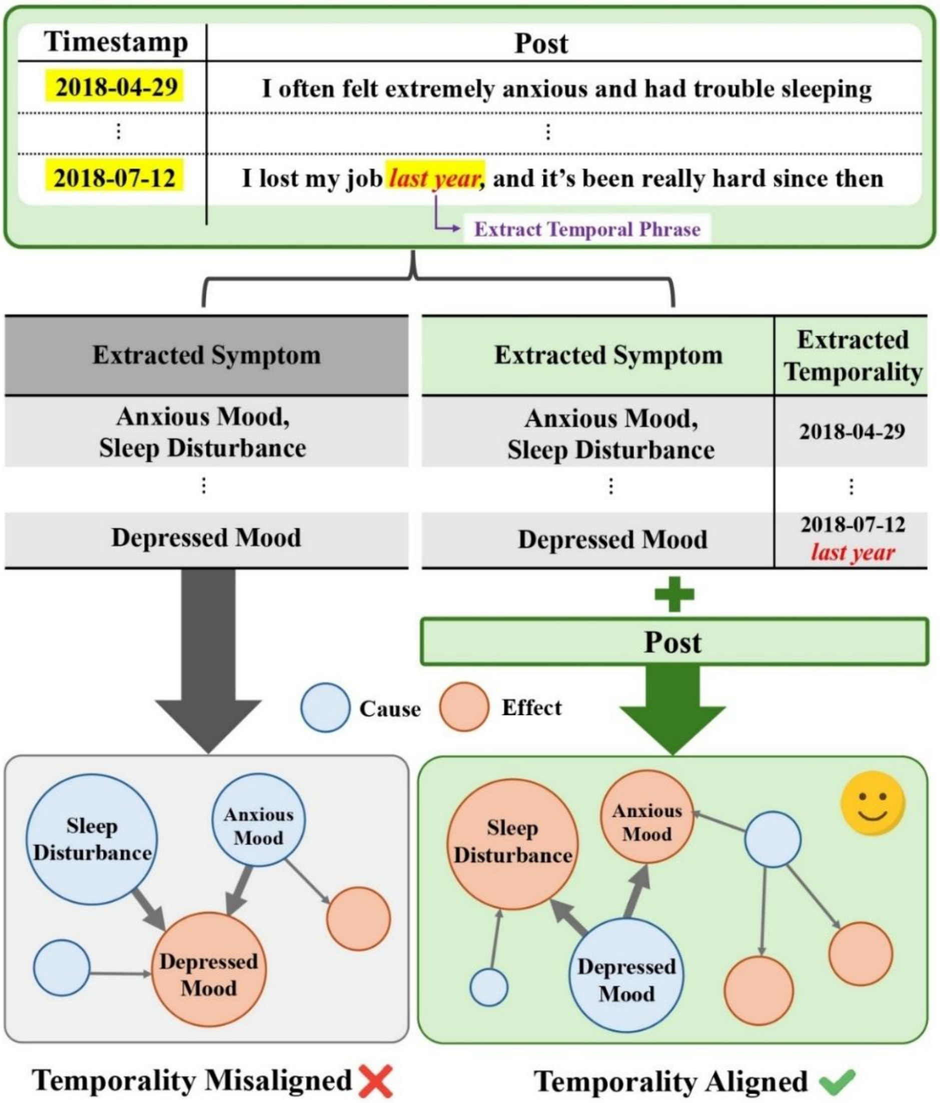

Figure 1. *Limitations of prior causal methods (e.g., ATE or PC-based) in modeling symptom progression. Prior methods ignore temporal expressions such as "last year," leading to incorrect causal directionality between symptoms. Our framework extracts such temporality using prompt-based LLMs and incorporates them to disambiguate the time order of symptom emergence, resulting in temporally coherent and more densely connected causal graphs.*

However, as illustrated in Figure 1, both ATE and PC-based methods neglect temporality, which is a crucial aspect of mental health data. Expressions like "last year" or a timestamp often convey how long a symptom persisted or when it emerged. Without accounting for these cues, causal models may infer misleading relations, such as interpreting an old, resolved symptom as the consequence of a recent one. Therefore, recent studies have begun exploring Large Language Models (LLMs) for causal discovery, leveraging their ability to capture latent semantics and event sequences from free-form text [6]. Yet, these approaches typically rely on static representations of time, often overlooking the diversity and subtlety of temporal expressions common in user-generated posts.

To overcome these limitations, we introduce TCPE (Temporal Causal Phrase Embedding), a novel framework for building causal graphs that reflect fine-grained symptom progression in mental health social media posts. Our method aligns LLM-based reasoning with explicit temporal phrases and timestamps extracted from user posts, allowing causal relations to reflect symptom evolution rather than static correlations. Extensive experiments on three public mental health datasets demonstrate that TCPE consistently improves both detection accuracy and graph quality, outperforming strong baselines such as ATE and PC. TCPE's integration of phrase-level temporal cues leads to more accurate identification of early-onset symptoms and more connected graphs centered on clinically important nodes. These results suggest that temporally grounded reasoning is key to unlocking the full potential of LLM−based causal discovery in mental health.

Chapter 2. Related Work

1. Causality in Mental Health

Causality is essential in mental health for identifying risk factors, designing interventions, and interpreting model decisions [7, 8]. However, randomized controlled trials (RCTs), while ideal, are often infeasible in this domain due to ethical and practical limitations.

As a result, observational methods such as the Average Treatment Effect (ATE) have been widely adopted, estimated via Propensity Score Matching (PSM) [9] or Inverse Probability Weighting (IPW) [10]. These techniques allow researchers to approximate randomized conditions by balancing covariates between treatment and control groups. In mental health research, such methods have been employed to estimate the effects of social media behaviors, life events, or therapy exposure on depressive symptoms [11, 12]. Recent work by Chen et al. [4] advances this line by constructing a pairwise symptom–symptom causal matrix using ATE. These pairwise scores are integrated with symptom features for downstream prediction tasks, showing improved performance on early depression detection benchmarks by aligning causal priors with neural representations.

Despite such progress, ATE-based methods are limited to predefined treatment-outcome pairs and produce only scalar effects, failing to capture multivariate dependencies or temporal symptom trajectories. To address this, structure learning approaches such as the PC (Peter-Clark) algorithm [13, 14] have been proposed to uncover causal graphs via conditional independence tests. While effective in low-dimensional, structured domains such as neuroimaging [15], these methods assume causal sufficiency and faithfulness—assumptions often violated in noisy, unstructured data like social media.

This motivates the need for more flexible frameworks that handle noisy, temporally misaligned symptom data without relying on strong assumptions or rigid pair definitions.

2. LLMs For Temporal Causal Reasoning

Large Language Models (LLMs) offer new opportunities to infer causality from structured narratives by leveraging contextual understanding. Prior studies have used LLMs for zero-shot causal discovery [16, 17], causal explanation [18], and counterfactual QA [19], primarily on formal or structured corpora [20]. In mental health, LLMs have also shown promise in tracking symptom trajectories [21], detecting early onset [22], and forecasting risks from user-generated posts [23, 24]. However, these models often overlook the semantics of temporal expressions (e.g., “since last month” or “after he left”), reducing them to surface cues rather than incorporating them into causal reasoning. While some methods attempt to encode temporal signals [25, 26], their focus remains on event ordering or temporal QA, not on modeling symptom interactions over time. Moreover, existing LLM−based causal approaches typically rely on tabular or structured data and are ill−suited for unstructured, noisy symptom narratives in social media.

In this work, we fill this gap by proposing a temporal−causal framework that explicitly conditions on time−anchored language. Our approach enables symptom−level causal reasoning grounded in naturalistic temporal expressions, enhancing interpretability and clinical relevance for mental health prediction.

Chapter 3. Research Design

1. Symptom and Text Marker Extraction

We begin by processing each user's timeline, which consists of dated social media posts. Each post undergoes a two-part extraction process.

Psychiatric Symptom Extraction

We adopt 38 psychiatric symptoms curated in the PsySym dataset [27], which covers seven major mental disorders and provides 83,000 annotated Reddit sentences. Symptoms such as *anxious mood*, *sleep disturbance*, and *suicidal ideation* are identified using BERT-based sequence classification models trained on this dataset. These clinically grounded symptom definitions serve as the foundation for causal reasoning in later stages.

Text Marker Extraction

In practice, mental health narratives are filled with subtle text markers, such as when a symptom started or how long it lasted. These markers are crucial for accurate causal inference, but conventional approaches either ignore them or reduce them to surface-level features.

To address this, we extract four types of text markers that frequently appear in user-generated content. These markers are designed to complement symptom labels by capturing event ordering, causality, persistence, and negation — all essential dimensions for modeling symptom interactions.

- **Temporal Markers:** Indicate timing or sequence (e.g., *"after I experienced"*, *"a few years ago"*).
- **Causal Connective Markers:** Explicitly express causation (e.g., *"because"*, *"due to"*, *"which results in"*).
- **Duration/Frequency Markers:** Represent persistence or intensity (e.g., *"constantly"*, *"for 7 months"*).
- **Negation Markers:** Indicate the absence of symptoms (e.g., *"not anxious anymore"*, *"never suicidal"*).

Markers are extracted using prompt-based GPT-4o queries with fixed output formats. The details of prompt designs can be found in Appendix 1. The outputs of this step are used to give additional clues for LLM-based causal discovery carried out in the next step.

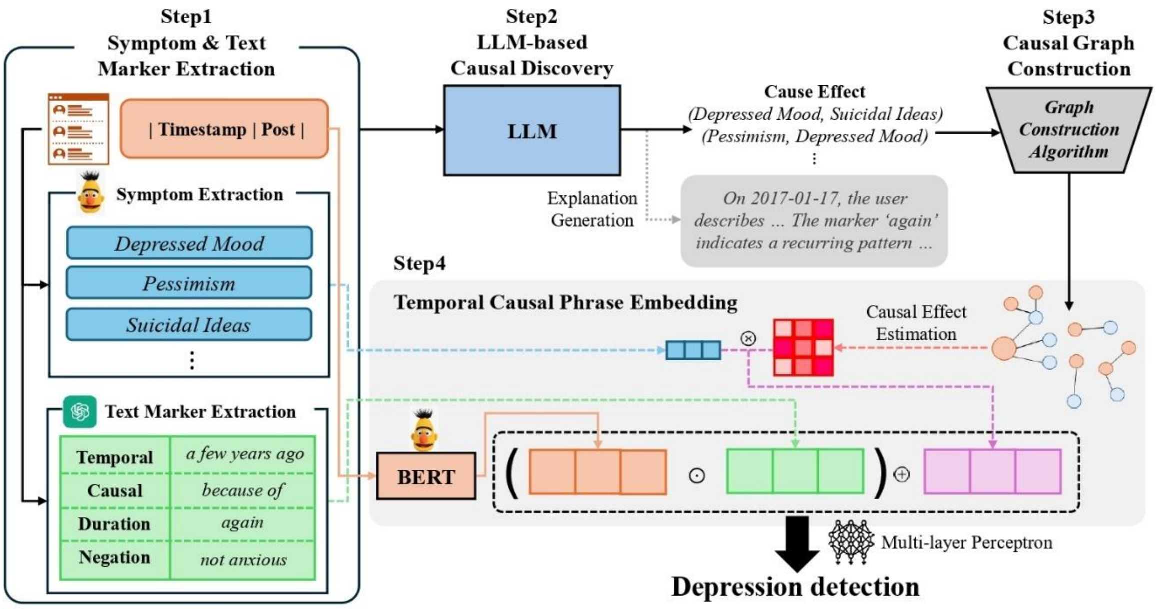

Figure 2. *Our proposed* ***TCPE (Temporal Causal Phrase Embedding)*** *framework consists of four sequential steps:* ***(1) Symptom & Text Marker Extraction:*** *From time-ordered social media posts, we extract psychiatric symptoms and text markers. (2)* ***LLM-based Causal Discovery:*** *A large language model (LLM) infers symptom-level causal pairs along with short natural-language explanations. (3)* ***Causal Graph Construction:*** *Candidate edges are filtered and composed into a cycle-free directed acyclic graph (DAG) using Algorithm 1 (Graph Construction with Statistical Filtering). (4)* ***Temporal-Causal Phrase Embedding:*** *Text markers are multiplied into contextual embeddings to enhance temporal–causal awareness for depression risk prediction.*

2. LLM−based Causal Discovery

Building on the extracted symptoms and text markers from the previous section, we leverage large language models (LLMs) to infer directional relationships between symptoms. Rather than treating user posts as isolated entries, our method constructs a structured prompt in which the user's timeline is arranged chronologically, and each post is annotated with both the extracted symptoms and their associated text markers. This design allows the LLM to ground its reasoning in the actual temporal order and contextual flow of symptoms, rather than relying on disconnected phrases.

The prompt used in this step presents a time−ordered sequence of posts enriched with symptom annotations and previously extracted text markers. It provides the model with the necessary narrative context to interpret how symptoms evolve and interact over time. Figure 3 shows the instruction section of the prompt, where the LLM is explicitly guided to output causal pairs in a structured format. The model is asked to identify potential *(cause, effect)* symptom pairs and to provide concise natural language explanations grounded in the content of the posts.

The output explanations can be found in Figure 2. Notably, the explanation incorporates explicit temporal references (e.g., dates or phrases like "On 2017−01−17") and text markers (e.g., "again") that serve as strong clues for inferring causal directionality. The details of the entire prompt design can be found in Appendix 2.

Rules
1. Prefer causal links between symptoms that appear on different days unless a same-day link is strongly supported by language.
2. For each pair, determine the lag (delay in days) between cause and effect based on their first appearances.
3. If the delay is expressed vaguely (e.g., “eventually”, “after a while”), estimate the lag.
4. Ignore negated symptoms (phrases in the Negation list) if they are clearly stated to be no longer present.
5. Do not invent new symptoms beyond the labeled ones. Only use clearly marked labels.

Explanation Guidelines
For each causal link, provide a detailed explanation that includes:
- Quotes or paraphrases from relevant posts
- The day-wise appearance of cause and effect (e.g., “Anxiety appeared on Day 1, while Insomnia first appeared on Day 3”)
- Supporting evidence from EXTRA CLUES (e.g., phrases like “since then”, “because I was anxious”)
- Avoid vague claims — base each link on explicit patterns or temporal reasoning.

Figure 3. *Instruction section of the prompt used for LLM-based causal discovery. The prompt guides the model to extract directional symptom pairs and provide evidence-grounded explanations.*

3. Causal Graph Construction

The raw output of LLM-based causal discovery is a set of directed symptom pairs $\mathcal{E}_{\mathrm{cand}} = \{(u, v)\}$, where u is a potential cause and v represents its effect. However, these outputs often contain noise, contradictions, or cycles. To obtain a reliable causal graph, we propose a post-processing algorithm that filters and validates edges in two steps. Let $\mathcal{M} \in \{0,1\}^{N \times d}$ denote the binary symptom matrix, where N is the number of users and d is the number of symptoms. Each entry $\mathcal{M}_{ij} \in \{0,1\}$ indicates whether user i has exhibited symptom j. The final output is $\mathcal{E}_{\mathrm{final}}$, representing the set of filtered causal edges.

Step 1: Statistical Edge Filtering

For each $(u, v) \in \mathcal{E}_{\mathrm{cand}}$, we first compute the joint frequency by counting how many users have exhibited both symptoms u and v in $\mathcal{M}$. If this frequency exceeds a predefined threshold θ_{freq}, we perform a chi-squared (χ^2) test between the binary presence vectors of u and v. Pairs with $p < 0.001$ are retained in $\mathcal{E}_{\mathrm{sig}}$, forming the set of statistically validated symptom pairs to be used in the next step.

Step 2: Cycle−Free Graph Construction

We then iteratively insert edges from $\mathcal{E}_{\text{sig}}$ into an initially empty Directed Acyclic Graph (DAG) G, ensuring that each added edge preserves acyclicity. If a reverse edge (v, u) already exists, a conflict resolution heuristic is applied to determine the correct direction. After processing all candidate edges, the retained edges are sorted in topological order to form the final $\mathcal{E}_{\text{final}}$.

This procedure is summarized in Algorithm 1, which outlines the statistical filtering and cycle−free graph construction process. The resulting structure $\mathcal{E}_{\text{final}}$ serves as the foundation for temporal modeling in the TCPE framework.

Algorithm 1: Graph Construction with Statistical Filtering

Input: Candidate edges from LLM: $\boldsymbol{\mathcal{E}}_{\text{cand}} = \{(\boldsymbol{u}, \boldsymbol{v})\}$,

Symptom matrix M $\in$ R$\boldsymbol{N}$ $\times \boldsymbol{d}$

Output: Filtered causal edges Efinal

Step 1: Statistical Edge Filtering

$\boldsymbol{\mathcal{E}}_{\text{sig}} \leftarrow \emptyset$

for each $(u, v) \in \boldsymbol{\mathcal{E}}_{\text{cand}}$ **do**

Compute joint_freq of $\{(u, v)\}$ active in M;

if *joint_freq* > $\boldsymbol{\theta}_{\text{freq}}$ then

Conduct $\chi 2$ test between columns $\boldsymbol{u}$ and $\boldsymbol{v}$;

if p−value < 0.001 **then**

Add (u, v) to $\boldsymbol{\mathcal{E}}_{\text{sig}}$;

Step 2: Cycle−Free Graph Construction

Initialize empty G;

for each $(u, v) \in \boldsymbol{\mathcal{E}}_{\text{sig}}$ **do**

if $(v, u) \in G$ **then**

Resolve directionality conflict;

if adding (u, v) keeps G acyclic **then**

Add (u, v) to G;

return $\boldsymbol{\mathcal{E}}_{\text{final}} = G$

4. Temporal Causal Phrase Embedding Modulation

To capture how temporal and causal expressions influence symptom-level dynamics, we introduce a modulation mechanism called **Temporal Causal Phrase Embedding (TCPE)**.

Input Representation

The TCPE modulation consists of three components. First, the merged text sequence of a user' s posts is tokenized and encoded using BERT. We extract the [CLS] embedding $h_{\text{cls}} \in \mathbb{R}^{H}$ as a holistic representation of the input, where H is the hidden size of the BERT model. Second, we use an adjacency vector $\text{adj\_vec} \in \mathbb{R}^{d}$ to represent the symptom-level causal structure, where d is the number of psychiatric symptoms described earlier. The learned causal graph is represented as a matrix $G_{\text{symp}} \in \mathbb{R}^{d \times d}$ (symptom × symptom), following the causal effect estimation strategy proposed by Chen et al. [4]. This matrix summarizes how a user' s observed symptoms causally interact. A user' s binary symptom vector $S_{\text{bin}} \in \{0,1\}^{1 \times d}$ is multiplied by the graph matrix to produce the final causal-symptom vector $\text{adj\_vec} = S_{\text{bin}} \cdot G_{\text{symp}} \in \mathbb{R}^{d}$, which encodes edge-weighted symptom interactions. Third, we construct a phrase vector $\text{phrase\_vec} \in \{0,1\}^{4}$, where each entry indicates the presence or absence of a specific text marker: temporal, causal, duration, or negation.

Modulation Mechanism

Given these inputs, the model computes a scaling vector $\gamma \in \mathbb{R}^H$ and a shifting vector $\beta \in \mathbb{R}^H$ from the phrase vector:

$$\gamma = 1 + \sigma(W_1 \cdot \text{phrase\_vec}), \beta = W_2 \cdot \text{phrase\_vec},$$

where $W_1, W_2 \in \mathbb{R}^{H \times 4}$ are trainable matrices, and $\sigma(\cdot)$ is the tanh function.

The modulated contextual embedding is defined as:

$$h_{\text{mod}} = \gamma \odot h_{\text{cls}} + \beta,$$

where $\odot$ denotes element-wise multiplication. This operation allows each dimension of the contextual embedding to be selectively amplified ($\gamma > 1$), attenuated ($\gamma < 1$), or shifted ($\beta \neq 0$) depending on which phrase types are present. Thus, temporal markers can enhance features encoding symptom progression, while causal markers emphasize dimensions relevant to cause-effect reasoning.

Integration with Causal Graph Features

Finally, the modulated embedding h_{mod} is concatenated with the adjacency causal vector adj_vec:

$$x = h_{\mathrm{mod}} \oplus \mathrm{adj\_vec},$$

where $\oplus$ denotes vector concatenation. This combined representation is passed through a feedforward classifier to compute logits. During training, the logits are compared with the ground-truth labels using BCEWithLogitsLoss to update the model parameters. During testing, the loss computation is omitted, and the logits are passed through a sigmoid function to obtain probabilities, which are then thresholded to generate binary predictions.

Chapter 4. Experiment

1. Baselines

To assess the impact of temporal–causal integration, we compare our framework (TCPE) with a diverse set of baselines grouped into three categories: ATE-based, Text-only, and Causal graph-based methods.

ATE-Based Model [4]

Pairwise Average Treatment Effects (ATE) between symptoms are computed using propensity score matching. The resulting vectors summarize indirect causal influence patterns and are concatenated with the h_{cls} embedding to serve as inputs to a binary classifier.

Text-Only Models

These baselines rely solely on raw post-level textual features without incorporating any symptom-level or causal information.

- BERT [28]: A pretrained BERT model is fine-tuned to classify users based on their concatenated post history. The h_{cls} embedding is used for binary classification through a linear layer.

- GPT−4o: GPT−4o receives a fixed prompt containing recent posts and directly outputs a binary decision using few−shot in−context learning.

Causal Graph−Based Models

These models incorporate structured symptom−to−symptom dependencies to enable more informed predictions.

- PC Algorithm [13]: A constraint−based causal discovery method applied to symptom variables. It iteratively tests conditional independence between symptom pairs to infer a causal graph representing potential causal pathways.

- TCPE (Ours): We extract symptom−level causal graphs by prompting GPT−4o on symptom pairs, followed by structured graph construction. These graphs are then conditioned on user posts using a temporal–causal embedding to generate symptom activations for classification.

All graph−based models (PC and TCPE) use the same underlying ATE matrix to assign edge weights, ensuring consistency across causal baselines.

2. Datasets

To evaluate the generalizability of our method, we use three large−scale, publicly available datasets that are widely adopted in mental health research.

eRisk2022 Dataset [29]

The eRisk dataset is part of the CLEF eRisk challenges, designed to facilitate early detection of depression through sequential analysis of Reddit user posts. We follow the dataset distribution used by Saeuberli et al. (2022), which includes 1,493 negative users with a total of 986,360 posts and 214 positive users with 90,222 posts.

Reddit Self−reported Depression Diagnosis (RSDD) Dataset [31]

The RSDD dataset contains Reddit posts from approximately 9,000 users who have self−reported a depression diagnosis, along with about 107,000 matched control users. Posts from mental health subreddits or those containing depression−related keywords were excluded to ensure that the analysis focuses on general, non−mental−health discussions.

Self-reported Mental Health Diagnoses (SMHD) Dataset [32]

The SMHD dataset consists of Reddit posts from users who have self-reported mental health conditions such as depression, anxiety, and bipolar disorder. It includes around 369,000 users, with approximately 83,000 self-reported cases and matched controls. In this work, we specifically use the subset of users labeled with depression.

3. Evaluation Metrics

To assess both predictive performance and causal fitness, we employ three evaluation metrics: **ERDE**, **Recall**, and **F1 Score**.

ERDE (Early Risk Detection Error) [33]

ERDE evaluates both the correctness and timeliness of predictions in early detection tasks. It assigns a full penalty to false negatives (FN), zero to true negatives (TN), and a decaying penalty to delayed true positives (TP) using a logistic function. False positives (FP) are also penalized in proportion to the class imbalance to discourage unnecessary early alarms.

Formally, ERDE is defined as:

$$\mathrm{ERDE}_{\phi} = \frac{1}{N}\sum_{i=1}^{M}\begin{cases} 1, & \text{if no alarm and } y_i = 1\text{(FN)} \\ c_{\phi}(p_i), & \text{if alarm at post } p_i \text{ and } y_i = 1\text{(TP)} \\ \text{fp\_p}, & \text{if } y_i = 0 \text{ and alarm(FP)} \\ 0, & \text{if } y_i = 0 \text{ and no alarm(TN)} \end{cases}$$

Here, $y_i$denotes the ground-truth label for user i, $N$is the total number of users, and $p_i$is the index of the post at which the first alarm was raised for user i. The delay cost is defined as $c_{\phi}(p_i) = 1 - \frac{1}{1+e^{(p_i-\phi)}}$, where $\phi \in \{5,50\}$controls how steeply the cost increases with delay. A smaller value of ϕ(e.g., 5) imposes a stronger penalty for delayed predictions. The false positive penalty, *fp_p*, is defined as the ratio of positive cases to the total number of users in the dataset.

Recall [34]

In mental health detection, recall is particularly important since missing true positive cases (false negatives) may result in delayed or missed interventions, leading to severe real-world consequences. We report Recall scores across all datasets to evaluate the sensitivity of each model.

F1 Score

The F1 score is computed at the user level by aggregating predictions across all posts of each user, thereby evaluating the overall detection quality per user. We report F1 scores across all datasets to ensure consistent and robust performance assessment.

4. Evaluation on Psychiatric Symptoms

To assess the reliability of symptom-level predictions, we evaluate individual symptom classification performance using a MentalBERT-based encoder [35] and a linear classifier. The model achieves a high average AUC of 0.994 and an average F1 score of 0.706 across 38 psychiatric symptoms, demonstrating strong overall discriminative performance despite class imbalance in underrepresented categories. We adopt the symptom taxonomy defined in the PsySym knowledge graph [27] to guide both symptom extraction and evaluation.

Such robust per-symptom classification provides a reliable foundation for downstream causal analysis, ensuring that the extracted symptom signals faithfully represent clinically meaningful patterns despite the inherent noise of social media posts. The performance metrics for each symptom are provided in Appendix 3.

5. Experiment Result

Table 1,2,3 summarizes the performance of all models on the eRisk, RSDD, and SMHD datasets using ERDE5, ERDE50, Recall, and F1. Across all three datasets, TCPE achieves the highest Recall and F1 scores, indicating that the integration of text markers significantly enhances overall classification accuracy. Although TCPE's ERDE values are moderately higher than those of the PC model, the consistent improvement in F1 demonstrates that incorporating text markers improves prediction quality, even at a slight trade-off in earliness. Among the causal baselines, PC obtains the lowest ERDE5 and ERDE50 across datasets, aligning with its strength in early detection through structure learning.

However, its lower F1 compared to TCPE suggests that structural inference alone is insufficient to handle label imbalance and semantic variability in user posts. This discrepancy and its implications are further discussed in the case study section (Section 6).

In addition, ATE performs better than text-only baselines in terms of F1 but still lags behind PC and TCPE, highlighting the limitations of scalar treatment–outcome models in capturing complex, multi-symptom trajectories. BERT and GPT-4o, when used with text-only inputs, show weaker performance across all metrics. This pattern indicates that relying solely on raw linguistic information without incorporating symptom-level structure or temporal context is inadequate for early mental-health risk detection.

Overall, the results demonstrate that TCPE consistently outperforms all baselines by achieving the highest Recall and F1 across datasets, confirming that integrating text markers provides the most accurate and reliable framework for depression detection in social media data.

Model	eRisk			
	ERDE5	ERDE50	Recall	F1
ATE	0.087	0.067	0.794	0.354
BERT	0.117	0.081	0.703	0.325
GPT-4o	0.136	0.092	0.651	0.275
PC	**0.086**	**0.066**	0.766	0.357
TCPE(Ours)	0.096	0.073	**0.842**	**0.403**

Table 1. *Performance of all compared models on the eRisk dataset using ERDE5, ERDE50, Recall and F1. Best scores are highlighted in bold. This table shows that methods incorporating symptom dependencies and temporal cues outperform other baselines, with TCPE achieving the highest Recall and F1-scores across all datasets.*

Model	RSDD			
	ERDE5	ERDE50	Recall	F1
ATE	0.093	0.069	0.782	0.342
BERT	0.115	0.079	0.698	0.321
GPT-4o	0.132	0.090	0.642	0.262
PC	**0.090**	**0.067**	0.741	0.345
TCPE(Ours)	0.094	0.072	**0.835**	**0.411**

Table 2. *Performance of all compared models on the RSDD dataset using ERDE5, ERDE50, Recall and F1. Best scores are highlighted in bold. This table shows that methods incorporating symptom dependencies and temporal cues outperform other baselines, with TCPE achieving the highest Recall and F1-scores across all datasets.*

Model	SMHD			
	ERDE5	ERDE50	Recall	F1
ATE	0.089	0.065	0.776	0.331
BERT	0.110	0.078	0.693	0.320
GPT-4o	0.125	0.087	0.639	0.260
PC	**0.087**	**0.063**	0.739	0.336
TCPE(Ours)	0.092	0.070	**0.850**	**0.417**

Table 3. *Performance of all compared models on the SMHD dataset using ERDE5, ERDE50, Recall and F1. Best scores are highlighted in bold. This table shows that methods incorporating symptom dependencies and temporal cues outperform other baselines, with TCPE achieving the highest Recall and F1-scores across all datasets.*

Chapter 5. Discussion

1. Ablation Study

A. Performance on LLMs

Table4,5 reports the performance of TCPE when paired with different LLMs during causal graph construction. We compare four models: Qwen-3 8B, Llama3-8B, Gemma-7B and GPT-4o. In all cases, we observe a clear improvement in F1 score when TCPE is applied, compared to the base performance of the LLMs used alone. The largest gains appear with GPT-4o, which achieves the highest F1 across all datasets when integrated with TCPE. This suggests that GPT-4o provides particularly compatible responses for causal prompting. TCPE consistently reduces ERDE scores as well, though the magnitude varies by model and dataset.

These results support two main observations. First, TCPE is adaptable to various LLM backbones, functioning effectively without relying on model-specific behaviors. Second, the LLM-based causal graph generation contributes meaningful structure to early-risk detection, regardless of which LLM generates the edges. This reinforces the general utility of our graph construction approach.

Model		Qwen-3 8B	TCPE (Qwen-3 8B)	Llama3-8B	TCPE (Llama3-8B)
eRisk	ERDE5	0.138	0.103	0.137	0.093
	ERDE50	0.094	0.075	0.093	0.072
	F1	0.270	0.374	0.272	0.379
RSDD	ERDE5	0.134	0.102	0.133	0.092
	ERDE50	0.091	0.074	0.090	0.071
	F1	0.258	0.366	0.259	0.371
SMHD	ERDE5	0.127	0.100	0.126	0.090
	ERDE50	0.088	0.072	0.087	0.071
	F1	0.255	0.359	0.257	0.362

Table 4. *Performance of TCPE combined with LLM backbones for causal discovery. TCPE consistently improves performance compared to text-only baselines, showing that its LLM-based graph generation generalizes beyond a single LLM.*

Model		Gemma-7B	TCPE (Gemma-7B)	GPT-4o	TCPE (GPT-4o)
eRisk	ERDE5	0.139	0.100	0.136	0.096
	ERDE50	0.095	0.073	0.092	0.073
	F1	0.268	0.370	0.275	0.403
RSDD	ERDE5	0.135	0.099	0.132	0.262
	ERDE50	0.092	0.072	0.090	0.072
	F1	0.256	0.363	0.262	0.411
SMHD	ERDE5	0.128	0.097	0.125	0.092
	ERDE50	0.089	0.070	0.087	0.070
	F1	0.253	0.357	0.260	0.417

Table 5. *Performance of TCPE combined with LLM backbones for causal discovery. TCPE consistently improves performance compared to text-only baselines, showing that its LLM-based graph generation generalizes beyond a single LLM.*

B. Ablation on Text−Derived Features

To examine how each linguistic feature contributes to overall model performance, we conduct a feature−level ablation study by selectively removing one component at a time from the phrase vector used during causal graph construction. Table 6 summarizes the resulting changes in early detection and classification metrics across datasets.

Among the four linguistic features, temporal expressions have the most significant influence. Removing them consistently reduces both ERDE and F1 scores, confirming their critical role in modeling how symptoms evolve over time. The removal of causal, negation, or duration expressions produces smaller but still measurable declines in performance, with duration features showing the least impact.

Overall, these findings demonstrate that temporal phrases act as essential anchors for constructing coherent and clinically meaningful causal relationships within users' posts, underscoring their importance for accurate early detection in mental health analysis.

Model		TCPE	w/o Temporal	w/o Causal	w/o Duration	w/o Negation
eRisk	ERDE5	0.096	0.101	0.098	0.097	0.097
	ERDE50	0.073	0.078	0.075	0.074	0.074
	F1	0.403	0.388	0.395	0.397	0.396
RSDD	ERDE5	0.094	0.099	0.096	0.095	0.095
	ERDE50	0.072	0.076	0.074	0.073	0.073
	F1	0.411	0.396	0.403	0.405	0.404
SMHD	ERDE5	0.092	0.097	0.094	0.093	0.093
	ERDE50	0.070	0.074	0.072	0.071	0.071
	F1	0.417	0.401	0.410	0.412	0.411

Table 6. *Impact of removing each text-derived feature from the phrase vector on model performance across datasets. The results show that temporal marker is the most critical feature.*

2. Analysis

A. LLM-as-a-Judge Evaluation [36,37]

To evaluate whether our prompt design improves the intuitiveness and clarity of causal explanations, we performed a pairwise comparison between explanations generated using the base prompt and those produced with the TCPE-enhanced prompt. For each comparison, both outputs were presented to GPT-4o, which was asked to judge which explanation better demonstrated causal reasoning.

As summarized in Table 7, explanations generated with the TCPE prompt were consistently preferred across all models. This finding indicates that incorporating text markers into the prompt structure leads to more coherent, logically grounded, and persuasive causal explanations. These results highlight the benefit of our approach for enhancing interpretability in mental health reasoning tasks. Details of the prompt designs are provided in Appendix 4.

Model	TCPE	Tie	Base Prompt
GPT-4o	**59.5 ± 2.1%**	0.9 ± 0.4%	36.9 ± 2.0%
Qwen3-8B	**63.8 ± 2.3%**	1.8 ± 0.6%	34.4 ± 2.2%
Llama3-8B	**65.9 ± 2.4%**	2.1 ± 0.7%	32.0 ± 2.1%
Gemma-7B	**61.7 ± 2.2%**	1.5 ± 0.5%	36.8 ± 2.3%

Table 7. *LLM-as-a-Judge evaluation between explanations generated by the Base prompt and the TCPE prompt. Results are reported as TCPE/Tie/Base Prompt percentages, with 95% confidence intervals.*

B. Analysis on Causal Graph

To evaluate the interpretability and clinical relevance of the constructed causal graphs, we compare the PC Algorithm with our TCPE method that integrates text markers (Table 8). While both graphs are constructed over the same set of symptoms, their structural properties differ significantly in terms of density and semantic richness.

The TCPE graph contains 19 edges, compared to 11 in the PC-derived graph, resulting in a higher average node degree (2.92 vs.1.65). This indicates that incorporating contextualized language cues enables the model to identify denser and more nuanced symptom interdependencies. In terms of semantic quality, 28.0% of the edges in TCPE satisfy the high-confidence criterion of having an ATE above 0.1, compared to 23.4% in the PC graph. This suggests that TCPE not only captures more relationships but also aligns

Of particular note is the difference in connectivity for two clinically critical symptoms: Depressed_Mood and Suicidal_ideas. The PC graph identifies 3 and 1 causal connections for these nodes, respectively, while the TCPE graph reveals 7 and 3 connections. Given that both symptoms are core diagnostic features of Major Depressive Disorder and are prominently represented in the PsySym ontology [27], the improved connectivity in TCPE underscores the importance of integrating text markers to recover clinically meaningful pathways that are often missed by structure-only approaches.

Metric	PC Algorithm	TCPE
# of Edges	11	19
Avg. Node Degree	1.65	2.92
ATE > 0.1	23.4%	**28.0%**
# of Edges for Depressed_Mood	3	**7**
# of Edges for Suicidal_ideas	1	**3**

Table 8. *Structural and semantic comparison between causal graphs generated using PC Algorithm versus TCPE.*

C. Case Study

While ERDE quantifies the timeliness of predictions in early detection settings, it does not fully account for semantic correctness, particularly in cases where models raise alarms on retrospective or resolved symptoms. In our experiments, PC obtains lower ERDE5 scores by triggering earlier, yet semantically incorrect, alarms. TCPE, in contrast, may receive higher penalties under ERDE but produces predictions that better match the actual onset and context of depressive symptoms.

To illustrate this discrepancy, we present two representative cases in Figure 4. Both User A and User B are labeled as depressed, yet the timing and rationale behind the predictions of PC and TCPE differ significantly. These examples show how temporal and causal markers help TCPE avoid false positives and better align predictions with ground-truth depression label.

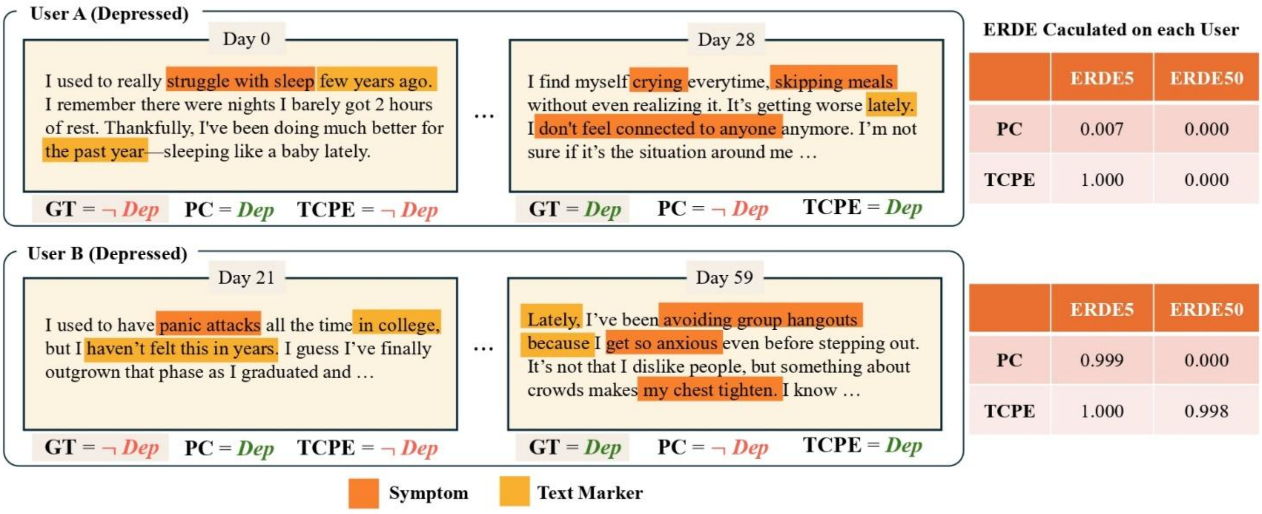

Figure 4. *Case study illustrating the limitations of ERDE in reflecting post-level ground truth. User A and User B both show depression. GT is the Ground Truth label of a post, and Dep denotes depressed, while ¬ Dep denotes not-depressed. Although PC achieves lower ERDE values by raising early alarms, it fails to align with actual symptom onset in user posts. In contrast, TCPE issues contextually grounded predictions that better match the temporal semantics of depression indicators.*

User A

PC raises an alarm on **day 0** based on a post describing severe sleep problems *"struggle with sleep"* in the past *"few years ago"* and noting improvement *"doing much better for the past year."* These temporal markers clearly indicate a resolved episode. PC, lacking explicit temporal modeling, misclassifies this as an active depression risk. In contrast, TCPE correctly defers prediction and issues its first alarm on **day 28**, when the user reports *"It's getting worse lately."* PC's early but incorrect prediction results in a low ERDE5 score (**0.007**), while TCPE's accurate yet slightly delayed detection receives the maximum ERDE5 penalty (**1.000**).

This discrepancy highlights a core limitation of the ERDE metric that it rewards earliness regardless of semantic correctness, failing to penalize temporally misaligned false positives.

User B

PC raises an alarm on **day 21** for a post reflecting on *"panic attacks all the time in college"* and explicitly stating *"but I haven't felt this in years."* The presence of depressive keywords, these temporal and negation markers clearly indicate a retrospective and resolved episode rather than an active one and misclassifies this as an ongoing risk. TCPE, by contrast, delays its first positive prediction until **day 59**, when the user writes *"Lately, I' ve been avoiding group hangouts"* and adds *"because I get so anxious"* with *"my chest tighten"* phrases that include causal connectives and present-tense indicators of current distress. These cues correctly signal an active episode and prompt a contextually grounded prediction. Although PC' s early but incorrect alarm yields a slightly lower ERDE5 score (0.999) compared to TCPE' s accurate but delayed detection (1.000), TCPE achieves a much higher ERDE50 score (**0.998 vs. 0.000**). This again underscores the limitation of ERDE as a timeliness-centric metric that rewards early alarms regardless of semantic correctness, failing to capture whether the prediction actually aligns with the ground-truth label.

Despite ERDE remaining as a standard metric for evaluating early risk detection, it penalizes semantically correct but delayed predictions more harshly than early but incorrect ones. Thus, relying solely on ERDE may misrepresent model quality, especially in domains like mental health, where misinterpreting symptom timing can lead to critical diagnostic errors. Incorporating temporal, negation, and causal markers improves prediction *precision* in early depression detection. TCPE avoids false positives caused by retrospective mentions and temporally negated symptoms, and issues alarms only when evidence of current distress is grounded in context.

Chapter 6. Conclusion & Future Work

In this paper, we introduced **TCPE** (Temporal Causal Phrase Embedding), a novel framework for constructing symptom-level temporal causal graphs from social media posts. By integrating LLM-based causal discovery with explicit text markers, TCPE captures fine-grained symptom progression that conventional ATE- or PC-based methods often miss. Extensive experiments on three public mental health datasets demonstrate that TCPE consistently achieves the highest F1-scores across datasets. Beyond improved predictive performance, TCPE produces interpretable causal graphs with richer connectivity around clinically important symptoms, facilitating more transparent decision support in mental health risk detection.

Looking ahead, TCPE offers promising avenues for extension. Although our study focused on depression, the framework can be naturally adapted to other disorders, where symptom trajectories and text markers may differ substantially. Another important direction is personalization. While the current causal graphs are constructed at the population level, future work could tailor graph structures to individual users' symptom histories, thereby better reflecting heterogeneous symptom pathways and improving early intervention strategies. By advancing both temporal reasoning and user-specific modeling, TCPE has the potential to become a generalizable, clinically meaningful tool for digital mental health.

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Appendix

<Appendix 1> Prompt Design for Marker Extraction

This appendix illustrates the detailed prompt structures used to extract four categories of linguistic markers from user-generated text: temporal, causal, duration/frequency, and negation markers. Each prompt was optimized to ensure consistent GPT-4o responses by enforcing a fixed JSON-like output format and explicit examples.

These marker extraction prompts enable the model to capture nuanced contextual cues such as event sequencing, causal reasoning, symptom persistence, and negated states that are not directly encoded in the symptom labels but are critical for constructing reliable symptom causal graphs

```
You are analyzing a user-written post to detect whether it expresses any
causal relationships between psychological symptoms, including those
with a time delay between cause and effect.

Instructions:
1. Focus on explicit causal connectives (e.g., "because", "due to", "as a
result", "leads to", "causes", "makes me feel").
2. If there is no causal connective, return an empty string.

Post:
"{post_text}"

Output Format:
"{{causal_marker}}"
Example: "because I felt anxious"
If none, return an empty string: ""
```

Figure 5. *Prompt design for causal connective marker extraction. This prompt extracts explicit causal connectors (e.g., “because,” “therefore,” “which leads to”) that indicate causal relationships between symptom events.*

You are analyzing a user-written post to detect whether it expresses any causal relationships between psychological symptoms, including those with a time delay between cause and effect.

Instructions:
1. Focus on temporal markers indicating delays between symptoms (e.g., "after", "before", "since", "for weeks", "eventually", "gradually").
2. If there is no temporal marker, return an empty string.

Post:
"{*post_text*}"

Output Format:
"{{temporal_marker}}"
Example: "after a week"
If none, return an empty string: "“

Figure 6. *Prompt design for temporal marker extraction. This prompt identifies time-related expressions (e.g., “after,” “recently,” “a few months later”) to capture the chronological sequence of symptom occurrences.*

You are analyzing a user-written post to detect whether it expresses any causal relationships between psychological symptoms, including those with a time delay between cause and effect.

Instructions:
1. Focus on duration and frequency markers
(e.g., "constantly", "frequently", "rarely", "for 3 months").
2. If there is no duration marker, return an empty string.

Post:
"{post_text}"

Output Format:
"{{duration_marker}}"
Example: "for three weeks"
If none, return an empty string: ""

Figure 7. *Prompt design for duration/frequency marker extraction. This prompt detects markers describing persistence, intensity, or frequency (e.g., "for weeks," "every day," "constantly"), which reflect symptom continuity or recurrence.*

You are analyzing a user-written post to detect whether it expresses any causal relationships between psychological symptoms, including those with a time delay between cause and effect.

Instructions:
1. Focus on negation markers
(e.g., "not anxious anymore", "no longer tired").
2. If there is no negation marker, return an empty string.

Post:
"{post_text}"

Output Format:
"{{negation_marker}}"
Example: "no longer feeling depressed"
If none, return an empty string: ""

Figure 8. *Prompt design for negation marker extraction. This prompt identifies expressions that negate symptom presence (e.g., "no longer depressed," "not anxious anymore") to accurately distinguish symptom remission or absence.*

<Appendix 2>Prompt Design for LLM−based Causal Discovery

This appendix presents the complete prompt template used for the LLM−based causal discovery stage. The prompt organizes a user’s timeline into a chronological list of posts, each annotated with extracted symptom labels and text markers from the previous stage. It then instructs the LLM to identify directional symptom relations by reasoning over the temporal and causal context provided.

The prompt begins with a structured instruction section, clearly defining the model’s role and output format. It specifies that the model should extract symptom pairs representing potential causal links (cause → effect) and provide concise natural language explanations grounded in the text. To ensure consistency and interpretability, outputs are formatted in a JSON−like schema with explicit separation between symptom pairs and justifications.

This design enables the LLM to leverage temporal dependencies, event ordering, and linguistic cues such as causal connectives or duration indicators when inferring symptom−level causal relations. It forms the foundation for constructing the temporal causal graph described in Section 3.3.

You are a mental health professional.
Below is a time-ordered sequence of posts written by one user, spanning multiple days.
Each post already carries the symptom labels detected for that day.

{timestamp, posts, symptoms}

Additional Clues For This User
The following phrases were automatically extracted from the user's text to help you detect time-delayed or causal relationships:

• Temporal markers (delay / order) ……… {temporal_markers}
• Causal connectives ………………………………… {causal_markers}
• Duration / frequency markers …………………… {duration_markers}
• Negation makers (symptom stopped) ……… {negation_markers}

Task – Time-aware Causal Analysis

You are analyzing a multi-day progression of symptom-labeled posts. Your goal is to detect cause → effect relationships between symptoms that may occur across different days.

Caution:

- Examine how symptoms appear, disappear, or evolve over time.
- Prefer causal links where the cause appears first and the effect emerges later, even if several posts apart.
- Use the provided EXTRA CLUES (temporal/cause markers, duration, negation) and the actual text to support your inferences.

Rules
1. Prefer causal links between symptoms that appear on different days unless a same-day link is strongly supported by language.
2. For each pair, determine the lag (delay in days) between cause and effect based on their first appearances.
3. If the delay is expressed vaguely (e.g., "eventually", "after a while"), estimate the lag.
4. Ignore negated symptoms (phrases in the Negation list) if they are clearly stated to be no longer present.
5. Do not invent new symptoms beyond the labeled ones. Only use clearly marked labels.

Explanation Guidelines
For each causal link, provide a detailed explanation that includes:
- Quotes or paraphrases from relevant posts
- The day-wise appearance of cause and effect (e.g., "Anxiety appeared on Day 1, while Insomnia first appeared on Day 3")
- Supporting evidence from Additional CLUES (e.g., phrases like "since then", "because I was anxious")
- Avoid vague claims — base each link on explicit patterns or temporal reasoning.

Figure 9. *Full prompt template for LLM-based causal discovery. This figure shows the entire instruction and input structure used to guide LLMs in identifying directional symptom relations. The prompt includes: (1) role and task description, (2) example input/output format, and (3) chronological user post data enriched with symptoms and text markers.*

<Appendix 3> Symptom Classification and Taxonomy

This appendix provides detailed visualizations of the per-symptom classification results and the full taxonomy of the 38 psychiatric symptoms used throughout this study.

Each symptom was classified using a MentalBERT-based encoder followed by a linear layer trained on multi-label annotations derived from the PsySym knowledge graph. The evaluation metrics include the Area Under the Curve (AUC) and F1 score for each symptom, reflecting both discriminative ability and balance between precision and recall.

Figure 10 shows the individual AUC and F1 results for all symptoms. The model demonstrates uniformly high AUC across most symptoms, with moderate variation in F1 due to differences in sample frequency and linguistic ambiguity within user posts.

Figure 11 lists the complete mapping of symptom IDs and their corresponding names as defined in the PsySym taxonomy, which ensures consistency between data annotation, model evaluation, and causal graph construction.

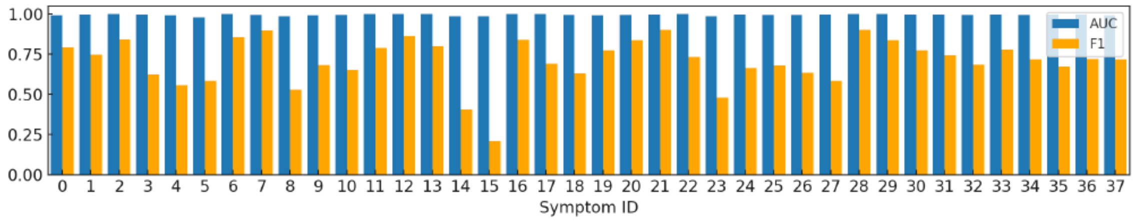

Figure 10. *Per-symptom classification performance (AUC and F1). Each bar represents the AUC and F1 score for one of the 38 psychiatric symptoms, showing robust discriminative power with consistent AUC values near 1.0.*

ID	Symptom	ID	Symptom	ID	Symptom
0	Anger Irritability	13	Respiratory symptoms	26	Intrusion symptoms
1	Anxious Mood	14	Suicidal ideas	27	Loss of interest or motivation
2	Autonomic symptoms	15	Worthlessness and guilty	28	More talkative
3	Cardiovascular symptoms	16	Avoidance of stimuli	29	Obsession
4	Catatonic behavior	17	Compensatory behaviors to prevent weight gain	30	Panic fear
5	Decreased energy tiredness fatigue	18	Compulsions	31	Pessimism
6	Depressed Mood	19	Diminished emotional expression	32	Poor memory
7	Gastrointestinal symptoms	20	Do things easily get painful consequences	33	Sleep disturbance
8	Genitourinary symptoms	21	Drastic shift in mood and energy	34	Somatic muscle
9	Hyperactivity agitation	22	Fear about social situations	35	Somatic symptoms others
10	Impulsivity	23	Fear of gaining weight	36	Somatic symptoms sensory
11	Inattention	24	Fears of being negatively evaluated	37	Weight and appetite change
12	Indecisiveness	25	Flight of ideas		

Figure 11. *Taxonomy of 38 psychiatric symptoms used in this study. The symptom list is adopted from the PsySym knowledge graph and forms the standardized label set used for both classification and causal analysis.*

<Appendix 4>Prompt Design for LLM-as-a-Judge Evaluation

This appendix presents the prompt template used for pairwise evaluation of causal explanation quality. In this setup, GPT-4o is instructed to compare two explanations. One is generated using the base prompt and another using our TCPE, enhanced prompt to select which better demonstrates causal reasoning. The structured format ensures consistent comparison criteria such as logical clarity, temporal grounding, and causal coherence.

```
You are comparing two explanations written by a mental health model about the causal
relationship between psychological symptoms.

Explanation 1:
"{explanation_1}"

Explanation 2:
"{explanation_2}"

Which explanation is better?

Criteria:
- Clarity of reasoning
- Specific connection to symptom cause-effect
- Reference to evidence in the original post
- Temporal logic (e.g., lag estimation)

Please answer with the number only:
0 → if it is a Tie
1 → if Explanation 1 is better
2 → if Explanation 2 is better
```

Figure 12. *Prompt template for LLM-as-a-Judge evaluation. This figure shows the full prompt used to evaluate causal explanation quality, where GPT-4o judges which of two model outputs exhibits stronger causal reasoning.*

<Appendix 5> Notation Summary

This appendix summarizes all mathematical symbols and variables used throughout the paper.

The notations cover the construction of symptom causal graphs, contextual modulation in the TCPE framework, and the evaluation metrics for temporal prediction. Each symbol is listed with its corresponding definition to ensure clarity and reproducibility of the proposed formulations.

Notation	Definition
$\mathcal{E}_{\text{cand}}$	Candidate edges from LLM
$\mathcal{E}_{\text{sig}}$	Statistically significant edges
$\mathcal{E}_{\text{final}}$	Final causal edges
G	Directed Acyclic Graph (DAG)
$\mathcal{M} \in \{0,1\}^{N \times d}$	Binary symptom matrix
N	number of users
d	number of symptoms
(u, v)	u = cause symptom , v = effect symptom
$h_{\text{cls}} \in \mathbb{R}^H$	BERT [CLS] embedding
$\texttt{adj\_vec} \in \mathbb{R}^d$	Causal-symptom vector
$\texttt{phrase\_vec} \in \{0,1\}^4$	Text marker vector
$G_{\text{symp}} \in \mathbb{R}^{d \times d}$	Symptom-level causal graph matrix
$S_{\text{bin}} \in \{0,1\}^{1 \times d}$	Binary symptom vector for a user
x	Final input vector to classifier
$\gamma, \beta \in \mathbb{R}^H$	Scaling and shifting vectors from TCPE
$W_1, W_2 \in \mathbb{R}^{H \times 4}$	Trainable weight matrices for TCPE
$\sigma(\cdot)$	tanh function
$h_{\text{mod}} \in \mathbb{R}^H$	Modulated contextual embedding
$\odot$	Element-wise multiplication
$\oplus$	Vector concatenation
ERDE_ϕ	Early Risk Detection Error
$c_\phi(p_i) = 1 - \frac{1}{1+e^{(p_i - \phi)}}$	Logistic penalty function for delay
$y_i \in \{0,1\}$	Ground-truth label for user i
p_i	Alarm-triggering post index
fp_p	false positive penalty

Figure 13. *List of mathematical notations and their definitions. This table provides a comprehensive summary of the variables and functions used in the model.*

논문요약

게시글로부터 시간적 인과관계로: 대형 언어모델을 이용한 증상 시간 인과그래프 구축

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정신건강 분석에서 기존 인과추론 기법(ATE, PC 알고리즘)은 증상의 시간적 변화와 상호작용을 충분히 반영하지 못하여 임상적 해석력이 제한되는 한계가 있습니다. 본 연구에서는 이를 해결하기 위해 대형 언어모델(LLM) 기반의 새로운 프레임워크인 TCPE(Temporal Causal Phrase Embedding) 를 제안하였습니다.

TCPE는 소셜미디어 게시글에서 정신건강 증상과 시간·인과 관련 표현(예: '최근', '~때문에', '매일', '이제는 아니다')을 추출하여 증상 간의 시간 인과그래프를 구축하는 방법입니다. GPT-4o를 활용한 프롬프트 설계를 통해 인과 방향성을 추론하고, 통계적 검정과 그래프 필터링을 수행하여 신뢰도 높은 인과 구조를 생성하였습니다. 또한 시간·인과 표현을 문맥 임베딩에 반영하여 증상의 진행 양상을 보다 정밀하게 모델링하였습니다.

eRisk, RSDD, SMHD 세 데이터셋을 활용한 실험 결과, TCPE는 기존 방법보

다 더 높은 F1 점수와 재현율을 달성하였으며, 특히 시간 표현을 제거할 경우 성능이 크게 하락하여 시간 정보의 중요성을 입증하였습니다. 또한 LLM 기반 설명 품질 평가 결과, TCPE 프롬프트가 더 일관적이고 설득력 있는 인과 설명을 생성하는 것으로 확인되었습니다.

결론적으로 TCPE는 언어 내 시간적 단서를 반영하여 정신건강 인과 구조를 정밀하게 모델링할 수 있으며, 조기 우울 탐지의 정확성과 해석 가능성을 동시에 향상시키는 프레임워크입니다. 향후에는 개인 맞춤형 인과그래프 확장 등을 통해 정밀 정신의학 분야로의 응용이 기대됩니다.

키워드: 인과추론, 정신건강, 시간적 인과임베딩, 소셜미디어 분석, 설명가능 인공지능